Same Behaviour, Different Mechanisms: Causal Layouts of Agreement Control Across Three Decoder-Only Language Models

Divyansh Agarwal

Abstract

A language model can produce the correct verb form without revealing how it got there. We ask whether three decoder-only models, Phi-2, Llama-3.2-3B, and Qwen2.5-3B, solve subject-verb agreement in the same way internally, using prompts where a singular or plural subject must control the verb despite an intervening noun, which we call the distractor. The attraction pattern is not the same across models: plural distractors disrupt Phi-2 and Qwen2.5-3B more than singular distractors do, while Llama-3.2-3B shows this asymmetry more weakly and less stably. We then use two causal interventions to compare the underlying control layout. Activation patching shows that agreement is driven mainly by the subject position rather than the distractor position in every model. Linear steering with a subject-number direction then separates the models: in Phi-2 and Llama-3.2-3B, the direction has a similar effect at the subject and final positions, with subject/final ratios of about $0.9\times$ and $1.0\times$, whereas in Qwen2.5-3B the effect stays concentrated at the subject, with a subject/final ratio of about $3.7\times$. The same ordering appears on held-out items, *was*/*were* prompts, eight anchor templates, all four subject-distractor conditions, and longer adverbial-filler prompts; effects sometimes shrink, but the model ordering does not reverse. We interpret the result as a difference in causal controllability rather than literal information flow: models that behave similarly can implement the same task through different internal causal layouts.

1 Introduction

Behavioural evaluations of language models measure what models output, not how they compute those outputs. A model can pass a controlled linguistic test while the causal organization supporting that behaviour remains unknown (Linzen et al., 2016; Marvin and Linzen, 2018). Models with comparable behavioural competence may therefore rely on different internal causal layouts. This matters for interpretability and evaluation: accuracy alone cannot establish mechanistic equivalence.

Subject-verb agreement is a useful testbed for this question. The task is well-defined, behaviourally measurable, and has a controlled syntactic structure with two diagnostic noun positions: the grammatical subject, which determines the correct verb number, and an intervening distractor noun, which may agree or conflict with the subject. Prior work has used agreement to study internal mechanisms in individual models (Lakretz et al., 2019; Finlayson et al., 2021), and recent work has analysed agreement circuits within a single model across languages (Ferrando and Costa-Jussà, 2024). What is missing is a *cross-model* causal comparison: do different language models that handle agreement comparably well organize that computation in the same way internally?

We address this question for three decoder-only models: Phi-2, Llama-3.2-3B, and Qwen2.5-3B. On controlled agreement prompts of the form "The {subject} of the {distractor} in question ___", all three models achieve high agreement accuracy and show distractor-attraction effects. However, their behavioural profiles are not identical: in Phi-2 and Qwen2.5-3B, plural distractors are more disruptive than singular distractors, whereas Llama-3.2-3B shows a weaker and more variable attraction pattern. Thus, all three models are competent on the task, but behavioural margins alone do not reveal whether they implement agreement through the same internal causal layout.

The causal interventions separate the models more clearly. Activation patching (Vig et al., 2020; Meng et al., 2022) shows that agreement control is primarily subject-driven rather than distractor-driven in all three models. Linear steering with a subject-number direction (Turner et al., 2023; Li et al., 2023) then reveals a model-level dissociation. In Phi-2 and Llama-3.2-3B, steering at the subject and final positions has comparable effect, with subject/final ratios of $0.88\times$ and $0.99\times$. In Qwen2.5-3B, the subject-position effect is $3.76\times$ stronger than the final-token effect, a pattern we call *source-localized*. The same subject/final ordering is seen again for *was/were*, held-out items, eight anchor templates, all four agreement conditions, and distance-stress prompts.

We frame these findings in terms of *causal controllability*: the degree to which adding a learned number direction to a token's hidden state changes the model's agreement decision. We do not claim that information literally moves between token positions; that stronger claim would require circuit-level attribution beyond the scope of this work.

Contributions.

- **Controlled behavioural comparison:** We construct controlled agreement prompts with four subject/distractor number conditions across three decoder-only models. All three models achieve high behavioural performance, while their attraction profiles are not identical (§3).
- **Subject-driven control:** Activation patching shows that agreement control is primarily subject-driven rather than distractor-driven in all three models, with subject/distractor patching ratios of about $6.6\times$ to $11.0\times$ in the main *is/are* paradigm (§4).
- **Cross-model causal dissociation:** Linear steering with a subject-number direction reveals that Phi-2 and Llama-3.2-3B are final-token integrated, with subject/final ratios near parity, about $0.9\times$ and $1.0\times$. Qwen2.5-3B is source-localized, with a much larger subject/final ratio of about $3.7\times$ (§5).
- **Robustness:** The subject/final ordering remains the same, although effect sizes sometimes shrink, for *was/were*, held-out items, eight anchor templates, all four agreement conditions, and adverbial-filler distance manipulations (§6).

2 Setup

2.1 Models

We study three publicly available decoder-only language models. Phi-2 (Javaheripi et al., 2023) has 32 transformer layers and uses partial rotary position embeddings. Llama-3.2-3B (Meta AI, 2024) has 28 layers and uses Rotary Position Embedding (RoPE) (Su et al., 2024). Qwen2.5-3B (Qwen Team, 2025) has 36 layers and uses RoPE with long-context scaling. All models are evaluated in inference mode without finetuning. Phi-2 is run in float32 for numerical stability; the other two are run in float16 on CUDA.

2.2 Prompt template and conditions

Our base template is

> *The* {subject} *of the* {distractor} *in question* ___

The blank position is where the model predicts the auxiliary verb. The subject determines the grammatically correct auxiliary number; the distractor is an intervening noun whose number may match or conflict with the subject. The phrase *in question* serves as a neutral anchor between the distractor and the verb slot; we test robustness to this anchor choice using eight alternative anchor templates in Section 6.

We construct four agreement conditions by independently varying subject and distractor number:

- **SS**: singular subject, singular distractor; correct auxiliary is singular.
- **SP**: singular subject, plural distractor; correct auxiliary is singular. This is the plural-attractor condition.
- **PS**: plural subject, singular distractor; correct auxiliary is plural. This is the singular-attractor condition.
- **PP**: plural subject, plural distractor; correct auxiliary is plural.

The first letter indicates subject number and the second indicates distractor number. The mismatch conditions SP and PS test whether an intervening distractor disrupts agreement with the true subject.

We use 31 noun pairs (e.g., *manager/managers*, *engineer/engineers*) and evaluate behavioural agreement across four auxiliary paradigms: *is/are*, *was/were*, *has/have*, and *does/do*. Causal experiments focus on the two be-auxiliaries (*is/are* and *was/were*), which give a clean present/past replication with single-token auxiliaries across all three models. After filtering to single-token auxiliaries and intersecting items across models, we retain 390 clean items for *is/are* and 392 for *was/were* in the causal experiments.

2.3 Behavioural measure

For each prompt x and model, we compute the verb-margin

$$\Delta(x) = \mathrm{logit}_x(v_{\mathrm{correct}}) - \mathrm{logit}_x(v_{\mathrm{incorrect}}),$$

where the correct auxiliary is determined by the subject's number. A positive value of $\Delta(x)$ means that the model assigns higher probability to the grammatically correct auxiliary than to the incorrect one.

Let $\mathcal{C} = \{\mathrm{SS}, \mathrm{SP}, \mathrm{PS}, \mathrm{PP}\}$ denote the set of agreement conditions, and let D_c denote the set of prompts in condition c. For each $c \in \mathcal{C}$, the condition-level mean verb-margin is

$$\Delta_c = \frac{1}{|D_c|} \sum_{x \in D_c} \Delta(x).$$

We summarize distractor attraction using two costs:

$$C_{\mathrm{plural}} = \Delta_{\mathrm{SS}} - \Delta_{\mathrm{SP}},$$

$$C_{\mathrm{singular}} = \Delta_{\mathrm{PP}} - \Delta_{\mathrm{PS}}.$$

Here, C_{plural} measures how much a plural distractor hurts a singular-subject prompt, while C_{singular} measures how much a singular distractor hurts a plural-subject prompt. Finally, we define attraction asymmetry as

$$A = C_{\mathrm{plural}} - C_{\mathrm{singular}}.$$

Positive values of A mean that plural distractors are more disruptive than singular distractors.

2.4 Causal interventions

Interventions at the subject, distractor, or final position target the contextual hidden state aligned with that position, not an isolated lexical representation. Our claims therefore concern where subject-number information is causally usable for the agreement decision.

We use two complementary causal methods. **Activation patching** swaps a hidden state at a chosen position and layer with the corresponding state from a donor prompt differing in the property being probed, and measures the resulting change in Δ (Vig et al., 2020; Meng et al., 2022). **Linear steering** (Turner et al., 2023; Li et al., 2023) adds a learned subject-number direction to a chosen position and measures the resulting change in Δ. In the main steering experiments, the steering strength is $\alpha = +1$, corresponding to movement in the plural-subject direction.

For both methods, we report bootstrap 95% confidence intervals over items. Controls include self-patching for patching, and shuffled-label plus random same-norm directions for steering. Self-patching should give zero effect; shuffled-label and random-direction controls should give near-zero effects.

The main steering experiment injects the direction at three diagnostic positions: subject, distractor, and final. We then extend the intervention to all prompt positions for a localization profile, with full position-by-layer heatmaps in Appendix E. Held-out splits, alternative anchors, and distance-stress variants test whether the dissociation survives direction overfitting, template changes, and increased intervening distance.

3 Behavioural results

Figure 1 shows the main *is/are* behavioural pattern. All three models maintain positive agreement margins across SS, SP, PS, and PP, indicating high behavioural competence. However, the attraction profiles differ: plural distractors reduce margins more strongly than singular distractors in Phi-2 (asymmetry 2.66) and Qwen2.5-3B (2.14), while Llama-3.2-3B shows a weaker asymmetry (0.28).

Appendix A reports the same analysis for *was/were*, *has/have*, and *does/do*. Accuracy remains high across paradigms. Phi-2 and Qwen2.5-3B show consistent positive plural-over-singular attraction asymmetry, while Llama-3.2-3B is weaker and more variable, including a small reversal for *does/do*.

These results show that all three models solve the task behaviourally, but the margins do not tell us which internal positions can causally control the agreement decision. The next two sections test this with causal interventions.

4 Patching: agreement control is subject-driven

We first localize the main causal control site for agreement. For each item, we construct paired prompts that differ in subject number or distractor number, then patch the contextual hidden state at a chosen position and layer.

Table 1 reports subject- and distractor-position patching effects for the pl→sg condition, where a plural-number donor state is patched into the cor-

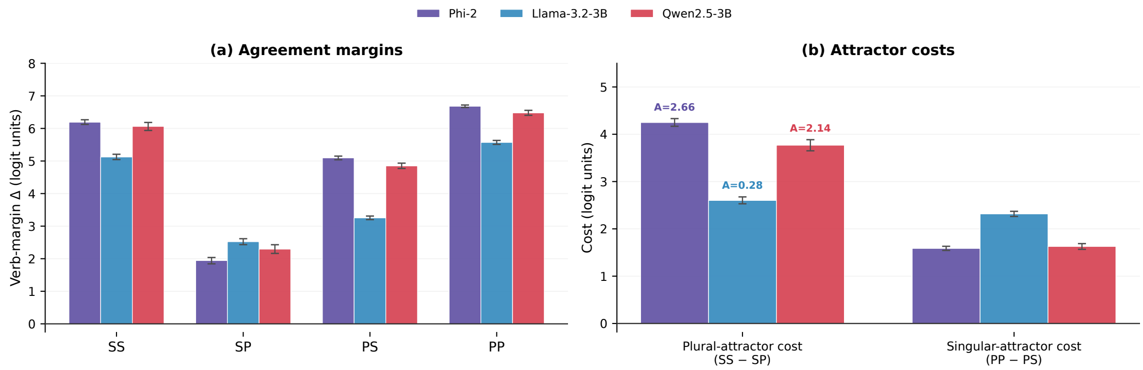

Figure 1: **Behavioural agreement performance and attraction asymmetry on the main *is/are* paradigm. (a)** Mean agreement margins across SS, SP, PS, and PP. All three models maintain positive margins. **(b)** Attraction-cost decomposition. The plural-attractor cost is SS − SP; the singular-attractor cost is PP − PS. Phi-2 (asymm. 2.66) and Qwen2.5-3B (2.14) show stronger plural-over-singular attraction asymmetry, while Llama-3.2-3B (0.28) shows a weaker asymmetry. Full behavioural results are provided in Appendix A.

Model	**Subj. patch**	**Dist. patch**	**Ratio**
is/are			
Phi-2	4.10	0.453	9.07×
Llama-3.2-3B	3.73	0.563	6.63×
Qwen2.5-3B	6.62	0.601	11.02×
was/were			
Phi-2	3.77	0.420	8.97×
Llama-3.2-3B	3.63	0.549	6.61×
Qwen2.5-3B	6.02	0.517	11.65×

Table 1: **Activation patching: subject position vs. distractor position.** Values are sign-oriented mean effects for the pl→sg patching condition, averaged over layers 1+. Positive values indicate movement toward the patched number. Subject-position patching is much stronger than distractor-position patching in all models and both be-auxiliary paradigms. Full bidirectional results are in Table 7; controls are in Table 8.

responding singular-number recipient prompt. Effects are sign-oriented, so positive values indicate movement in the expected patched-number direction. Subject/distractor ratios are large in all three models, ranging from 6.63× to 11.02× for *is/are* and from 6.61× to 11.65× for *was/were*. The reverse sg→pl direction also shows larger subject-position than distractor-position effects, though with smaller ratios (Table 7). Self-patching gives exactly zero, and crossed-distractor checks are much smaller than the main subject-position effects (Table 8).

Thus, agreement control is primarily subject-driven in all three models. The next section asks where this subject-number control is causally usable for the final prediction.

5 Steering: cross-model dissociation

5.1 Method

We build a subject-number direction separately for each layer:

$$\mathbf{d}^{(\ell)}_{\text{subj}} = \bar{h}^{\text{plural}}_{\ell,\text{subj}} - \bar{h}^{\text{singular}}_{\ell,\text{subj}},$$

where the means are computed over hidden states at the subject position for plural-subject and singular-subject prompts, respectively. Because $\mathbf{d}^{(\ell)}_{\text{subj}}$ is plural-minus-singular, positive signed effects indicate movement toward the plural auxiliary, not improved grammatical accuracy; larger effects mean greater causal usability at that position.

We evaluate the main steering analysis on SP, the lowest-margin behavioural condition, where a singular subject conflicts with a plural distractor. Section 6 then tests the same localization pattern on held-out items, *was/were*, anchor variants across all four conditions, and distance stress.

We inject at the **subject**, **distractor**, and **final** positions, average signed effects over layers 1+, and exclude the embedding layer. We use $\alpha = +1$; Appendix D reports the $\alpha = -1$ sign check and shuffled-label/random same-norm controls.

5.2 Main result

Table 2 and Figure 2 report the main result. In Phi-2 and Llama-3.2-3B, the subject-number direction is causally usable at both the subject and final positions, with subject/final ratios near parity in both be-auxiliary paradigms: 0.88× and 0.99× for *is/are*, and 0.87× and 0.97× for *was/were*.

In Qwen2.5-3B, the same direction has a strong effect at the subject position (4.83 logit units) but

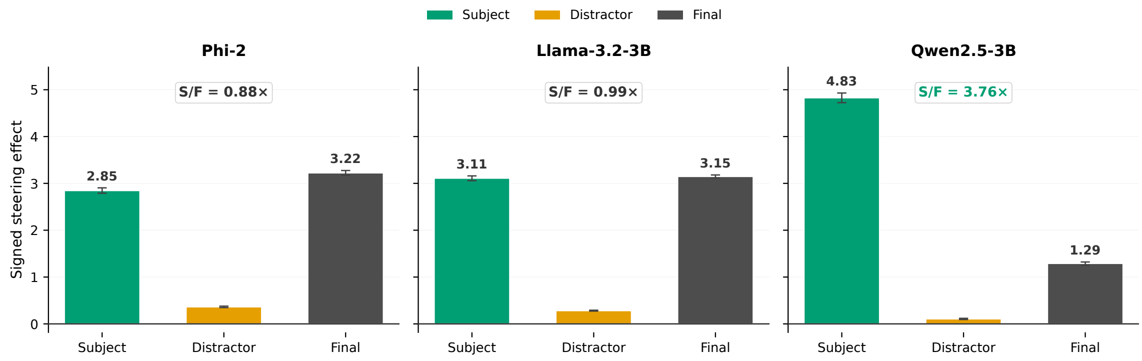

Figure 2: **Cross-model steering profile (main result).** Signed steering effect, averaged over layers 1+, when the subject-number direction is added at each of three positions, with shared y-axis. **Phi-2** and **Llama-3.2-3B** show subject and final effects of comparable magnitude. **Qwen2.5-3B** shows a strong subject-position effect (4.83) and a much weaker final-position effect (1.29), yielding a subject/final ratio of 3.76×. Distractor-position effects are weak in all models.

Model	Subj.	Dist.	Final	S/F
is/are				
Phi-2	2.85	0.36	3.22	0.88×
Llama-3.2-3B	3.11	0.28	3.15	0.99×
Qwen2.5-3B	4.83	0.11	1.29	**3.76**×
was/were				
Phi-2	2.76	0.35	3.19	0.87×
Llama-3.2-3B	3.11	0.28	3.20	0.97×
Qwen2.5-3B	4.46	0.09	1.20	**3.72**×

Table 2: **Steering effects at three positions.** Signed effects on the agreement margin, averaged over layers 1+. Controls are near zero relative to real steering effects, with maximum absolute effect 0.063. The subject/final ratio (S/F) reveals the cross-model dissociation: Phi-2 and Llama show similar effects at subject and final positions, while Qwen's subject-position effect is 3.76× stronger than its final-position effect for *is/are* and 3.72× stronger for *was/were*.

a substantially weaker effect at the final position (1.29 logit units), yielding a subject/final ratio of 3.76×. The pattern replicates closely for *was/were*, with a ratio of 3.72×. Distractor-position steering is small in all models, consistent with the patching result that the distractor position is not a strong causal control site.

We call the Phi-2/Llama pattern *final-token integrated*: the subject-number direction is causally usable at the prediction position. We call the Qwen pattern *source-localized*: causal usability is concentrated at the source, namely the subject position, with limited usability at the final position. Thus, behaviourally competent models differ in where subject-number control is causally usable.

5.3 All-position profile

To map the layout in finer detail, we extend the same SP steering setup to every token position (*The* / subject / *of* / *the* / distractor / *in* / final). Figure 3(a) shows the resulting profile per model.

The profile gives three useful checks. First, the subject region carries substantial causal usability in all models, including the subject position itself, and immediately following contextual positions such as *of*. This reflects contextual carry-forward under causal self-attention rather than an isolated lexical property of any function word. Second, all three models show weak causal usability at the distractor position: even when distractor number conflicts with subject number, distractor steering is much smaller than subject or final steering. Third, the cross-model dissociation visible in Figure 2 is part of a broader temporal pattern: in Qwen, the subject and adjacent post-subject positions retain high causal usability across much of the network, whereas in Phi-2 and Llama causal usability rises sharply at the final token in mid-to-late layers. Full position-by-layer heatmaps are in Appendix E.

5.4 Temporal dynamics

To quantify the temporal asymmetry visible in the all-position trajectories, Table 3 reports two summary statistics per model and verb paradigm. The source/final ratio divides subject-region AUC (summed subject and immediately post-subject effects across layers 1+) by final-token AUC. The **final-onset layer** is the first layer where the layer-wise final-token effect exceeds 1.5 logit units. Full AUC values are in Appendix C.

Model	*is*/*are*		*was*/*were*	
	Src/Final	**Onset**	**Src/Final**	**Onset**
Phi-2	$1.80\times$	15	$1.76\times$	15
Llama-3.2-3B	$1.75\times$	13	$1.73\times$	13
Qwen2.5-3B	$\mathbf{5.96}\times$	**28**	$\mathbf{5.93}\times$	**28**

Table 3: **Temporal dynamics of causal controllability.** Src/Final is the source/final AUC ratio, where source AUC sums subject-region effects across layers $1+$ and final AUC sums final-token effects. Onset is the first layer where the layerwise final-token effect exceeds 1.5 logit units. Qwen has a much larger source/final ratio and a later final-token onset in both verb paradigms. Full AUC values are reported in Appendix C.

Both statistics replicate closely across paradigms. Qwen has a source/final ratio of $5.96\times$ (*is*/*are*) and $5.93\times$ (*was*/*were*), compared with $1.80\times/1.76\times$ for Phi-2 and $1.75\times/1.73\times$ for Llama. Its final-token onset is layer 28 in both paradigms, versus layer 15 for Phi-2 and layer 13 for Llama. Alternative thresholds (1.0, 2.0) preserve this pattern, with Qwen's onset delayed by at least 13 layers.

These dynamics support a *handoff-like* interpretation in Phi-2 and Llama, where causal controllability builds up at the final position over mid-layers, and a *prolonged source-side control* pattern in Qwen, where the subject region retains controllability across most of the network and the final-token representation becomes a strong steering target only in late layers. We frame this as a description of where the steering direction is causally usable across layers, not as evidence of literal token-to-token information movement or as identification of the circuit components that implement the pattern.

6 Robustness

We test whether the cross-model dissociation depends on choices in the main experiment. Figure 3, panels (b) and (c), summarises the result visually; here we report each check in turn.

6.1 Held-out items

We rebuild $\mathbf{d}_{\text{subj}}^{(\ell)}$ from a randomly held-out 50% of items and test steering on the other half. Subject/final ratios change by less than 0.06 in all three models and both verb paradigms; the direction is not overfit to specific items. Full effects are reported in Appendix F.

Distance	Phi-2	Llama	Qwen
medium	$0.45\times$	$0.59\times$	$\mathbf{1.48}\times$
long	$0.55\times$	$0.62\times$	$\mathbf{1.78}\times$
extra long	$0.68\times$	$0.69\times$	$\mathbf{1.90}\times$

Table 4: **Distance-stress subject/final ratios.** The dissociation is preserved across all distances; in Qwen, the ratio grows with distance.

6.2 Verb paradigm

was/*were* ratios (Table 2) differ from *is*/*are* by at most 0.05 in any model. The same subject/final ordering appears for both be-auxiliary paradigms.

6.3 Anchor templates

To isolate template-level robustness, we hold the auxiliary pair fixed to *is*/*are* and vary only the anchor phrase. This asks whether the subject/final localization pattern depends on the original *in question* template; auxiliary-pair robustness is tested separately by the *was*/*were* steering, held-out, all-position, and temporal analyses. Across eight anchor phrases and all four agreement conditions, subject/final ratios are $0.57\times$ (Phi-2), $0.61\times$ (Llama-3.2-3B), and $1.33\times$ (Qwen2.5-3B). Thus, template variation compresses absolute effects but preserves the cross-model dissociation. Full anchor results, including the SP-only breakdown, are in Appendix G.

6.4 Distance stress

We insert noun-free adverbial fillers (e.g., "quite clearly, rather obviously, as expected,") between the distractor and the verb, and report three non-short buckets; the short condition is excluded because the distractor and final prediction positions coincide. Prepositional phrases are avoided to prevent additional attractors.

Subject and final steering effects compress with distance, but the dissociation persists at every bucket (Figure 3(b) and Table 4). Qwen's ratio actually *grows* with distance ($1.48\times \rightarrow 1.90\times$), while Phi-2 and Llama remain below parity. Distance therefore does not make the three models converge to the same subject/final ratio.

An all-position decomposition under distance stress confirms this is not driven by signal spreading into filler tokens: filler effects remain small (≤ 0.78 at medium, ≤ 0.32 at extra long) and shrink with distance, while the cross-model ordering at subject and final positions is preserved. Full decompositions are reported in Appendix H.

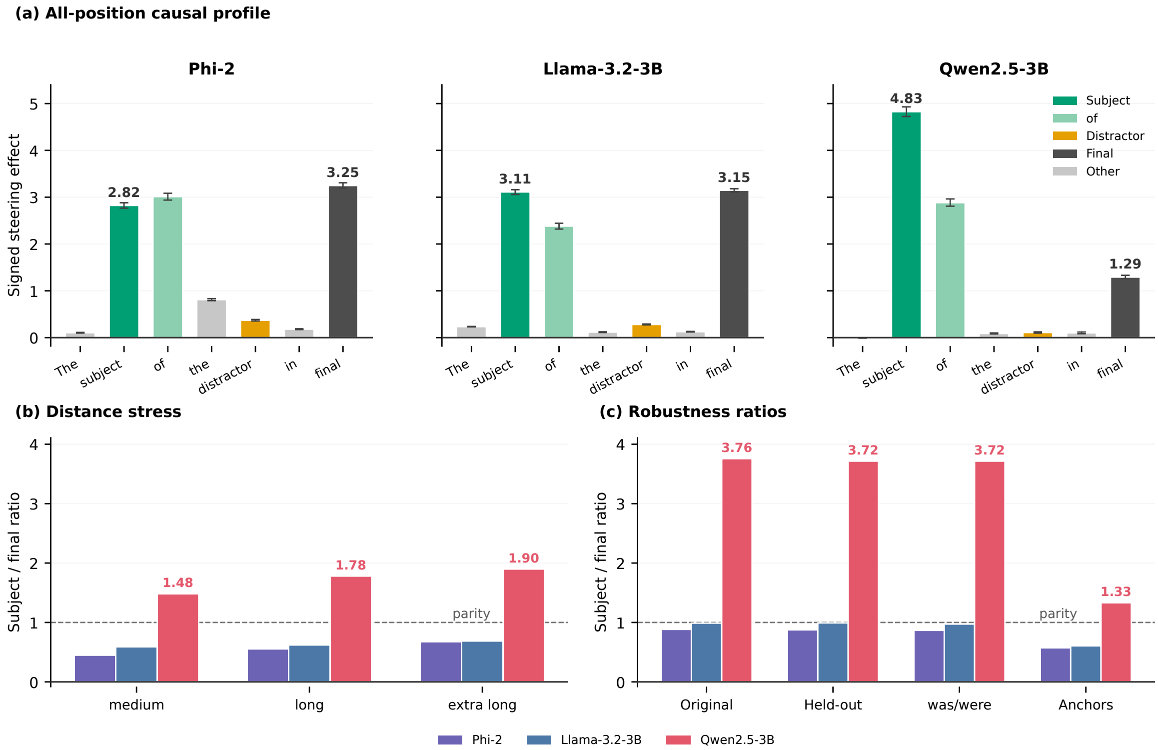

Figure 3: **Localization and robustness of the cross-model dissociation. (a)** All-position causal profile, averaged over layers 1+. Subject-adjacent effects reflect contextual carry-forward under causal self-attention; distractor-position effects remain weak. **(b)** Subject/final steering ratio across three distance buckets. Effects compress with distance in all models, but the dissociation is preserved: Phi-2 and Llama remain below parity, Qwen remains above. **(c)** Subject/final steering ratio across held-out items, the *was/were* replication, and eight anchor templates. The dashed line in (b) and (c) marks parity between subject and final controllability.

Check	Change	Result
Held-out	50% split	max change ≤ 0.06
was/were	verb pair	max change ≤ 0.05
Anchors	8 endings	preserved
Conditions	SS/SP/PS/PP	preserved
Distance	adverbial fillers	attenuated, no reversal

Table 5: **Robustness summary.** Each row varies one aspect of the setup and reports whether the model-level dissociation is preserved. The cross-model pattern survives every stress condition tested. Full held-out, anchor-template, and distance-stress results are reported in Appendices F, G, and H.

6.5 Summary

Table 5 summarises the robustness checks. Across these checks, Phi-2 and Llama remain closer to final-token control, while Qwen remains more subject-localized. None of the tested changes reverses this ordering.

7 Discussion

7.1 Interpretation and scope

Three decoder-only models that achieve comparable behavioural accuracy on subject-verb agreement implement that capability through measurably different internal causal layouts. Patching identifies a shared property: agreement control is subject-driven in all three models. Steering then identifies a divergence: in Phi-2 and Llama-3.2-3B the subject-number direction is causally usable at both the subject and final prediction positions, whereas in Qwen2.5-3B it is concentrated at the subject position and substantially weaker at the final position. The pattern is robust to the stress conditions we tested.

7.2 Implications

These implications should be read in terms of causal controllability rather than literal information flow. We characterize *where* the subject-number direction is causally usable in the network, not how the model arrived at that arrangement. Steering measures controllability: adding the direction at a position changes the agreement margin. It does not directly demonstrate that information is literally routed between token positions, nor does it identify which attention heads, MLP layers, or path-level routes implement the dissociation. Such circuit-level attribution is an important direction for future work, but it is separate from the model-level dissociation demonstrated here.

Under this interpretation, the cross-model difference matters for evaluation and interpretability. If two models with comparable behavioural competence implement a task through different internal causal layouts, then behavioural evaluation alone cannot detect the difference. Passing a behavioural test does not certify that two models share an internal mechanism, and it does not establish that an intervention which works on one model will transfer to another. Likewise, claims about "the agreement circuit" in one model do not automatically generalize, even within the same task and the same model-family scale.

7.3 Future work

Extending this analysis to larger model families and controlled architectural comparisons could test whether the source-localized versus final-token-integrated distinction generalizes beyond the 2 to 3B scale. Circuit-level attribution, in particular attention-head and MLP-level analyses, could identify which components mediate the final-token integration seen in Phi-2 and Llama and why Qwen2.5-3B exhibits prolonged source-side controllability.

8 Related work

Agreement in language models. Subject-verb agreement has been a standard testbed since Linzen et al. (2016), with extensive behavioural studies in LSTMs and transformers (Marvin and Linzen, 2018; Lakretz et al., 2019; Newman et al., 2021). Causal studies have identified agreement-related circuits in individual models (Finlayson et al., 2021; Ferrando and Costa-Jussà, 2024).

Causal interpretability methods. Activation patching (Vig et al., 2020; Meng et al., 2022) and linear steering / activation addition (Turner et al., 2023; Li et al., 2023; Zou et al., 2023; Rimsky et al., 2024) are now standard tools in mechanistic interpretability. Circuit-level analyses (Wang et al., 2022; Conmy et al., 2023; Nanda et al., 2023; Merullo et al., 2024) extend these to head- and MLP-level attribution.

Cross-model comparisons. Most causal interpretability work targets a single model. Recent work has probed circuit generality along adjacent axes: across tasks (Merullo et al., 2024), across languages (Ferrando and Costa-Jussà, 2024), and in toy settings across trained networks (Chughtai et al., 2023). Cross-model causal comparison of the same task remains underexplored. We study this missing comparison directly: for the same English agreement task, we test whether different model families expose the same causal layout. They do not.

9 Conclusion

We presented a cross-model causal analysis of subject-verb agreement in three decoder-only language models. All three models pass the behavioural test, with high accuracy and broadly similar, but not identical, attraction behaviour. Causal interventions reveal that they do not implement the task the same way internally. Phi-2 and Llama-3.2-3B expose strong final-token controllability; Qwen2.5-3B is source-localized, with steering effects concentrated near the subject. The dissociation is robust across verb pairs, held-out items, anchor templates, agreement conditions, and distance manipulations. We argue that comparable behavioural performance does not imply mechanistic equivalence, and that cross-model causal evidence is needed to make claims about how language models implement even simple linguistic tasks.

Limitations

Our study covers three decoder-only models at the 2 to 3B parameter scale. We therefore do not claim that the source-localized versus final-token-integrated distinction is a universal taxonomy of language models. Architectural differences, tokenizer choices, positional encoding schemes, and training data are correlated in our model set and cannot be disentangled here. We treat Qwen2.5-3B's source-localized profile as an empirical cross-model difference rather than as evidence for any single architectural cause.

The dissociation is robust across the stress tests we evaluate: held-out items, the *was*/*were* paradigm, anchor-template variation, all four agreement conditions, and distance manipulations. However, we do not identify the circuit-level cause of the difference. The observed source-versus-final controllability pattern is therefore evidence about where the steering direction is causally usable, not a full account of the attention heads, MLP layers, or path-level routes that implement agreement.

Ethics Statement

This work analyzes publicly released language models using synthetic subject-verb agreement prompts. It does not involve human subjects, private data, or sensitive demographic inference. The main risk is over-interpreting causal interventions as complete circuit explanations; we therefore frame the results as causal controllability rather than literal information flow or component-level attribution. Code and artifacts are released to support independent checking; see Appendix I for the repository link and reproducibility details.

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A Behavioural results across auxiliary paradigms

Table 6 reports full behavioural margins, accuracies, and attraction measures for all four auxiliary paradigms: *is/are*, *was/were*, *has/have*, and *does/do*. Because this appendix reports behavioural breadth rather than causal intervention results, we use each model's clean behavioural set after tokenization filtering; the all-model intersections used for causal experiments are reported in Section 2. For each model and verb pair, we report mean margins for SS, SP, PS, and PP, plural-attractor cost $\mathrm{SS}-\mathrm{SP}$, singular-attractor cost $\mathrm{PP}-\mathrm{PS}$, and asymmetry $(\mathrm{SS}-\mathrm{SP})-(\mathrm{PP}-\mathrm{PS})$. On the retained clean items, accuracy is 1.00 in all $4\times3\times4=48$ cells.

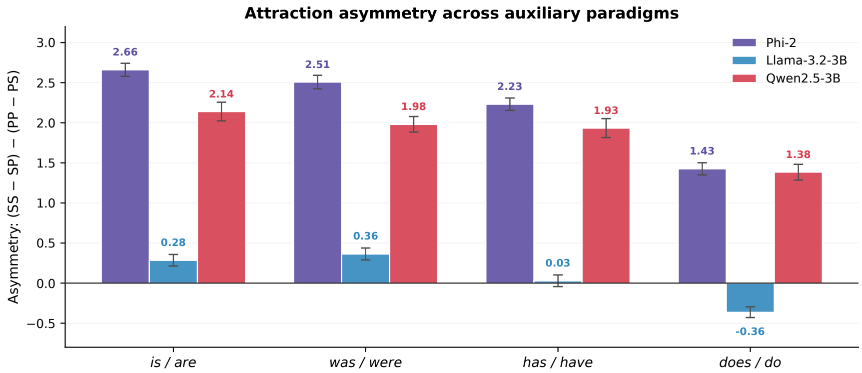

Figure 4: **Behavioural attraction asymmetry across all auxiliary paradigms.** Bars show $(\mathrm{SS}-\mathrm{SP})-(\mathrm{PP}-\mathrm{PS})$ for each model and verb pair, with bootstrap 95% CIs. Positive values indicate stronger plural-over-singular attraction. Phi-2 and Qwen2.5-3B show consistently positive asymmetry across *is/are*, *was/were*, *has/have*, and *does/do*; Llama-3.2-3B shows weaker and more variable asymmetry, including a small reversal in *does/do*. Full condition-level margins are reported in Table 6.

Figure 4 plots attraction asymmetry across all models and auxiliary paradigms. Phi-2 and Qwen2.5-3B show consistently positive plural-over-singular attraction asymmetry across the four paradigms. Llama-3.2-3B shows weaker and more variable asymmetry, including a small negative value for *does/do*. Importantly, accuracy remains high across paradigms, so this behavioural variability does not undermine the causal comparison; rather, it motivates the intervention-based analysis of where agreement control is causally usable.

Verb pair	SS	SP	PS	PP	Pl. cost	Sg. cost	Asymm.	N
Phi-2								
is/are	6.19	1.94	5.09	6.68	4.25	1.59	2.66	399
was/were	5.87	1.92	4.76	6.20	3.95	1.44	2.51	400
has/have	5.40	1.80	5.36	6.72	3.59	1.36	2.23	396
does/do	3.98	1.44	5.19	6.31	2.54	1.12	1.43	371
Llama-3.2-3B								
is/are	5.12	2.52	3.25	5.57	2.60	2.32	0.28	399
was/were	4.92	2.40	3.42	5.59	2.53	2.16	0.36	399
has/have	4.48	2.37	3.47	5.55	2.11	2.08	0.03	399
does/do	3.59	2.12	3.40	5.23	1.46	1.83	−0.36	399
Qwen2.5-3B								
is/are	6.06	2.29	4.85	6.48	3.77	1.63	2.14	391
was/were	5.50	2.12	4.40	5.80	3.38	1.40	1.98	392
has/have	5.73	2.39	5.03	6.44	3.34	1.40	1.93	386
does/do	4.74	2.12	4.85	6.09	2.62	1.24	1.38	382

Table 6: **Full behavioural agreement results across auxiliary paradigms.** For each model and verb pair, we report mean verb-margins for SS, SP, PS, and PP; plural-attractor cost $\mathrm{SS}-\mathrm{SP}$; singular-attractor cost $\mathrm{PP}-\mathrm{PS}$; asymmetry $(\mathrm{SS}-\mathrm{SP})-(\mathrm{PP}-\mathrm{PS})$; and number of clean items. Accuracy is 1.00 in every cell on the retained clean items.

B Patching: full details

Table 7 reports full bidirectional activation-patching results for both be-auxiliary paradigms. The main paper reports the pl→sg direction because it gives the clearest subject/distractor separation; the reverse sg→pl direction shows the same qualitative pattern, with subject-position patches larger than distractor-position patches in all three models.

Table 8 reports self-patching, crossed-distractor checks, and an additional subject-position check. Self-patching is exactly zero because the hidden state is replaced with itself. Crossed-distractor effects are nonzero because the substituted hidden states are contextual, but they remain much smaller than the main subject-patching effects. The SS→PP subject-position patch is not a null control; its effects are larger than crossed-distractor effects in all three models and both verb paradigms.

Layer-by-layer patching curves and item-level results are provided in the supplementary code repository.

Verb pair	Model	Patch direction	Subj. patch	Dist. patch
is/are	Phi-2	sg→pl	1.98	1.282
	Phi-2	pl→sg	4.10	0.453
	Llama-3.2-3B	sg→pl	2.21	0.838
	Llama-3.2-3B	pl→sg	3.73	0.563
	Qwen2.5-3B	sg→pl	4.33	1.432
	Qwen2.5-3B	pl→sg	6.62	0.601
was/were	Phi-2	sg→pl	1.87	1.189
	Phi-2	pl→sg	3.77	0.420
	Llama-3.2-3B	sg→pl	2.24	0.818
	Llama-3.2-3B	pl→sg	3.63	0.549
	Qwen2.5-3B	sg→pl	3.95	1.287
	Qwen2.5-3B	pl→sg	6.02	0.517

Table 7: **Full bidirectional activation-patching results.** Values are mean signed effects on the agreement margin, averaged over layers $1+$. The main paper reports the pl→sg direction because it gives the clearest subject/distractor separation, but the reverse sg→pl direction also shows larger subject-position than distractor-position effects in all models and both verb paradigms.

Verb pair	Model	Self SS	Self SP	Cross SP→PS	Cross SS→PP	Subj. SS→PP
is/are	Phi-2	0.000	0.000	0.234	0.386	1.226
	Llama-3.2-3B	0.000	0.000	0.362	0.670	2.016
	Qwen2.5-3B	0.000	0.000	0.403	0.591	4.015
was/were	Phi-2	0.000	0.000	0.231	0.340	1.153
	Llama-3.2-3B	0.000	0.000	0.357	0.608	1.910
	Qwen2.5-3B	0.000	0.000	0.350	0.500	3.584

Table 8: **Self-patching, crossed-distractor checks, and an additional subject-position check.** Self-patching gives exactly zero because the hidden state is replaced with itself. Crossed-distractor effects are nonzero because the substituted hidden states are contextual, but they remain much smaller than the main subject-patching effects reported in Table 1 and Table 7. The SS→PP subject-position patch is not a null control; it provides an additional check that subject-position representations carry causally important number information. Its effects are larger than crossed-distractor effects in all three models and both verb paradigms.

C Full temporal dynamics

Model	Src AUC	Final AUC	Src/Final	Onset L.
is/are				
Phi-2	180.9	100.7	1.80×	15
Llama-3.2-3B	148.3	85.0	1.75×	13
Qwen2.5-3B	269.8	45.2	**5.96×**	**28**
was/were				
Phi-2	175.8	99.6	1.76×	15
Llama-3.2-3B	148.9	86.3	1.73×	13
Qwen2.5-3B	249.9	42.2	**5.93×**	**28**

Table 9: **Full temporal-dynamics summary.** Src AUC sums subject-region effects across layers $1+$. Final AUC sums final-token effects. Src/Final is their ratio. Onset layer is the first layer where the final-token effect exceeds 1.5 logit units.

Table 9 reports the underlying source-side and final-token AUC values that feed into the source/final ratios in the main paper (Table 3). Source-side AUC sums signed steering effects across layers $1+$ over the subject region. Final AUC sums signed effects at the final token. Onset layer is the first layer where the final-token effect exceeds 1.5 logit units.

D Steering: controls and alpha sweep

Table 10 reports shuffled-label and random same-norm controls for the subject-number steering direction. The shuffled-label control preserves the direction-construction procedure but breaks the association with subject number, while the random same-norm control tests whether a vector of comparable magnitude can reproduce the effect. For each model and verb pair, we report the largest absolute control effect across the subject, distractor, and final injection positions.

Verb pair	**Model**	**Max** $\lvert\text{shuf}\rvert$	**Max** $\lvert\text{rand}\rvert$
is/are	Phi-2	0.006	0.063
	Llama-3.2-3B	0.003	0.011
	Qwen2.5-3B	0.002	0.020
was/were	Phi-2	0.004	0.036
	Llama-3.2-3B	0.006	0.032
	Qwen2.5-3B	0.007	0.027

Table 10: **Steering control summary.** Values are the largest absolute shuffled-label or random same-norm control effect across subject, distractor, and final injection positions for the subject-number direction, averaged over layers $1+$. Controls are near zero compared with the real steering effects reported in Table 2.

The controls are small in every case: the largest shuffled-label effect is 0.007, and the largest random same-norm effect is 0.063. These values are far below the real steering effects in Table 2, showing that the main intervention depends on the learned subject-number direction rather than vector norm alone.

Table 11 reports the final-token alpha sign check for the subject-number direction. Values are raw changes in the agreement margin Δ before the sign-orientation used in the main steering tables. In the SP condition, Δ favours the singular auxiliary, while the learned direction is plural-minus-singular. Therefore, adding the direction with $\alpha = +1$ should decrease Δ, because it pushes the model toward the plural auxiliary. Conversely, adding the opposite direction with $\alpha = -1$ should increase Δ, because it pushes the model back toward the singular auxiliary. This is exactly what we observe in all three models and both be-auxiliary paradigms.

Model	$\alpha = -1$ **raw effect**	$\alpha = +1$ **raw effect**
is/are		
Phi-2	+3.79 [+3.75, +3.82]	−3.22 [−3.27, −3.17]
Llama-3.2-3B	+2.96 [+2.92, +3.00]	−3.15 [−3.18, −3.12]
Qwen2.5-3B	+2.38 [+2.35, +2.41]	−1.28 [−1.32, −1.25]
was/were		
Phi-2	+3.71 [+3.67, +3.75]	−3.19 [−3.24, −3.14]
Llama-3.2-3B	+2.84 [+2.80, +2.88]	−3.20 [−3.23, −3.17]
Qwen2.5-3B	+2.15 [+2.12, +2.18]	−1.20 [−1.23, −1.17]

Table 11: **Alpha sign check for subject-number steering.** Values are raw final-token effects on the agreement margin Δ with bootstrap 95% CIs. In the SP condition, the plural-minus-singular direction decreases Δ at $\alpha = +1$ and increases it at $\alpha = -1$, confirming the expected direction orientation.

The alpha sweep is used as an orientation check, not as a separate localization metric. The magnitudes need not be perfectly symmetric because the intervention is applied inside a nonlinear model, but the consistent sign reversal verifies that the steering direction has the intended orientation.

E All-position steering heatmaps

Figure 5 shows the full position-by-layer steering heatmaps for the main *is/are* all-position experiment. These heatmaps provide the layer-resolved version of the layer-averaged profile shown in Figure 3(a). They show that Phi-2 and Llama develop strong final-token controllability in mid-to-late layers, while Qwen retains stronger source-side controllability and shows a later final-token onset.

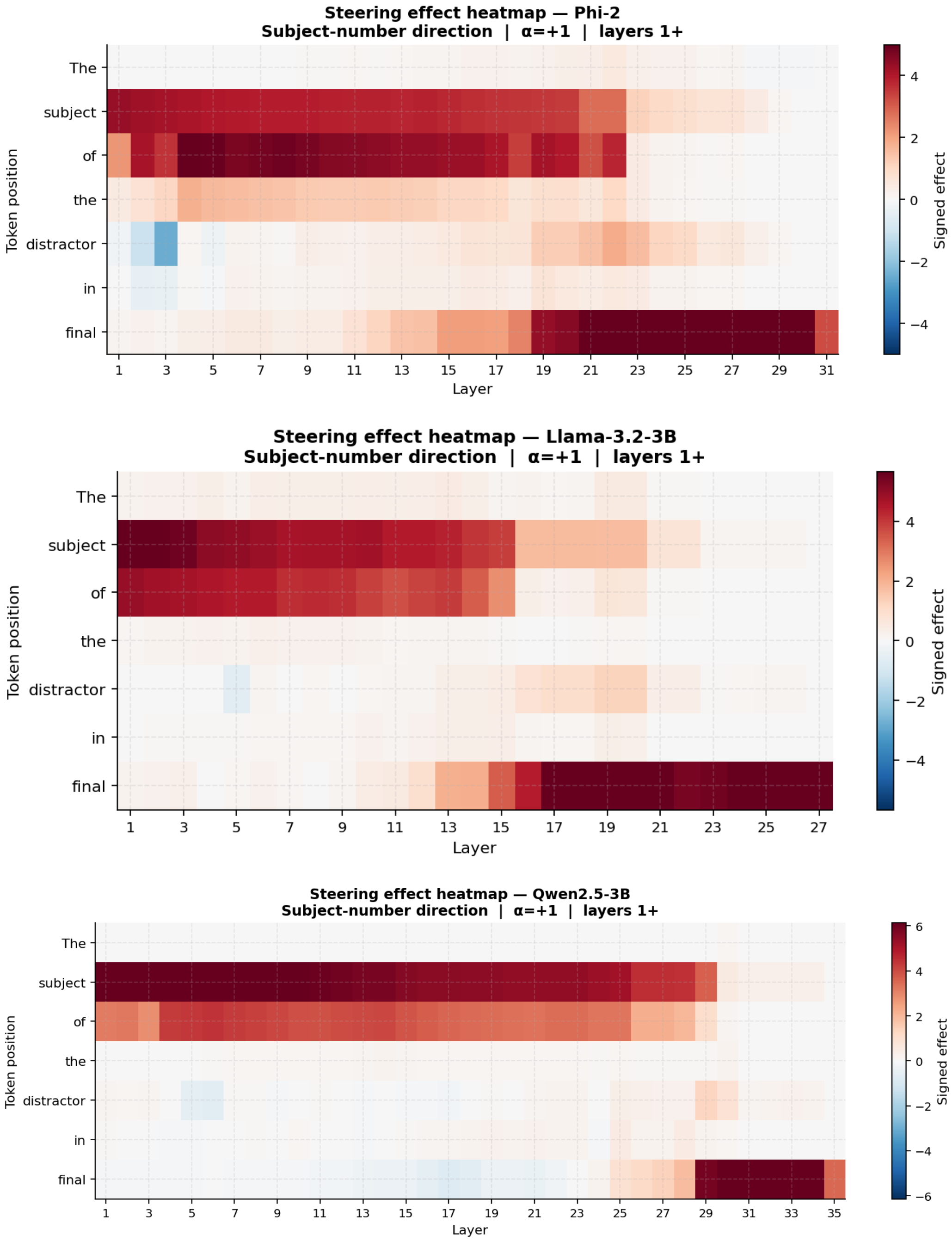

Figure 5: **Position-by-layer all-position steering heatmaps.** Each heatmap shows the signed steering effect of the subject-number direction across token positions and layers, excluding the embedding layer. Top: Phi-2. Middle: Llama-3.2-3B. Bottom: Qwen2.5-3B. These heatmaps provide the layer-resolved counterpart to the layer-averaged all-position profile in Figure 3(a).

F Held-out steering

Table 12 reports held-out steering results for both be-auxiliary paradigms. For each model, the subject-number direction is trained on one half of the items and evaluated on the held-out half. The subject/final ratios closely match the main steering results, showing that the dissociation is not driven by item-level overfitting.

Model	Subj.	Dist.	Final	S/F
is/are				
Phi-2	2.85	0.37	3.26	0.88×
Llama-3.2-3B	3.16	0.29	3.19	0.99×
Qwen2.5-3B	4.88	0.12	1.31	**3.72**×
was/were				
Phi-2	2.71	0.35	3.19	0.85×
Llama-3.2-3B	3.13	0.29	3.22	0.97×
Qwen2.5-3B	4.53	0.10	1.21	**3.74**×

Table 12: **Held-out steering results.** The subject-number direction is trained on a random 50% split and evaluated on held-out items. Values are signed steering effects averaged over layers 1+. The subject/final ratio remains close to the main result in both verb paradigms, indicating that the dissociation is not an artifact of direction overfitting.

G Anchor robustness

This analysis is a template-level robustness check. We intentionally hold the auxiliary pair fixed to *is/are*, because the goal is to isolate whether changing the anchor phrase affects the subject/final localization pattern. Auxiliary-pair robustness is evaluated separately in the *was/were* steering, held-out, all-position, and temporal analyses.

The eight anchor phrases used are: *in question*, *under review*, *on duty*, *on site*, *at work*, *in charge*, *under observation*, and *on call*. The steering direction is trained on the original *in question* template and evaluated on the anchor-variant prompts. Behavioural accuracy at the four-condition level remains above 0.99 in all three models.

Model	Subj.	Dist.	Final	S/F
Phi-2	1.88	0.67	3.28	0.57×
Llama-3.2-3B	1.66	0.49	2.74	0.61×
Qwen2.5-3B	2.70	0.58	2.03	**1.33**×

Table 13: **Anchor-template robustness for *is/are* across all four agreement conditions.** Values are signed steering effects averaged over layers 1+ and over the eight anchor templates. S/F is the subject/final ratio. The auxiliary pair is held fixed so that this analysis isolates template variation. The dissociation is preserved despite changing the anchor phrase.

Table 13 reports anchor robustness across all four agreement conditions. Effect sizes compress under template variation, but the dissociation is preserved: Phi-2 and Llama remain final-integrated, while Qwen remains source-localized.

Model	Subj.	Dist.	Final	S/F	S/D
Phi-2	2.99	0.57	3.45	0.87×	5.21×
Llama-3.2-3B	2.83	0.41	2.84	0.99×	6.97×
Qwen2.5-3B	4.52	0.18	1.67	**2.70**×	24.84×

Table 14: **Anchor-template robustness for *is/are* in the SP condition.** The SP condition has a singular subject and plural distractor. Holding the auxiliary pair fixed isolates whether the anchor phrase affects the localization pattern. The Qwen source-localized pattern remains especially clear, while Phi-2 and Llama remain close to final-integrated. S/D is the subject/distractor ratio.

Table 14 reports the same anchor-template evaluation restricted to the critical SP condition. The SP-only result shows the same pattern more sharply: Qwen has a much larger subject/final ratio than Phi-2 and Llama, and distractor-position steering remains weak.

H Distance experiment

We use four distance buckets:

- **short** (no filler): "The manager of the restaurants"
- **medium**: + ", quite clearly,"
- **long**: + ", quite clearly, rather obviously, as expected,"
- **extra long**: + ", quite clearly, rather obviously, as expected, at least for now, in most cases,"

The short distance condition is omitted from position-specific comparisons because the distractor token is also the final prediction token. The remaining buckets measure how subject-side and final-token controllability change as noun-free material is inserted between the distractor and the verb.

Model	Distance	Subject	Between	Filler	Final	S/F
Phi-2	medium	2.42	1.76	0.77	5.38	0.45×
	long	2.26	1.47	0.60	4.09	0.55×
	extra long	1.81	0.97	0.30	2.68	0.68×
Llama-3.2-3B	medium	1.88	0.69	0.55	3.18	0.59×
	long	2.04	0.78	0.48	3.29	0.62×
	extra long	1.60	0.62	0.32	2.33	0.69×
Qwen2.5-3B	medium	3.99	1.60	0.78	2.69	**1.48×**
	long	3.86	1.46	0.44	2.17	**1.78×**
	extra long	3.13	1.27	0.23	1.65	**1.90×**

Table 15: **All-position decomposition under distance stress.** Values are signed steering effects averaged over layers $1+$. S/F is the subject/final ratio. Filler-token effects are small and decrease with distance, while the cross-model ordering at subject and final positions is preserved.

Table 15 reports the all-position decomposition under distance stress. Subject and final effects attenuate with distance in all models, but the model-level ordering is preserved. Filler-token effects remain small and shrink with distance, indicating that attenuation is not simply due to the steering signal spreading into filler tokens.

Across the six token roles (prefix / subject / between / distractor / filler / final), effects attenuate from medium to extra long at every meaningful position; bootstrap $p < 0.001$ for all 18 tested cells in the *is/are* distance experiment.

I Compute and reproducibility

All experiments use bootstrap 1000-2000 resamples over items for confidence intervals. Phi-2 is evaluated in float32; Llama-3.2-3B and Qwen2.5-3B are evaluated in float16. Hidden states are extracted with `output_hidden_states=True`; steering is implemented with a forward hook on the relevant decoder layer. We use HuggingFace Transformers, PyTorch, NumPy, Pandas, and Matplotlib. Random seeds, cleaned item subsets, processed result CSVs, figure scripts, and evaluation code are released in the anonymous repository: https://anonymous.4open.science/r/same-behaviour-different-mechanisms-1716.